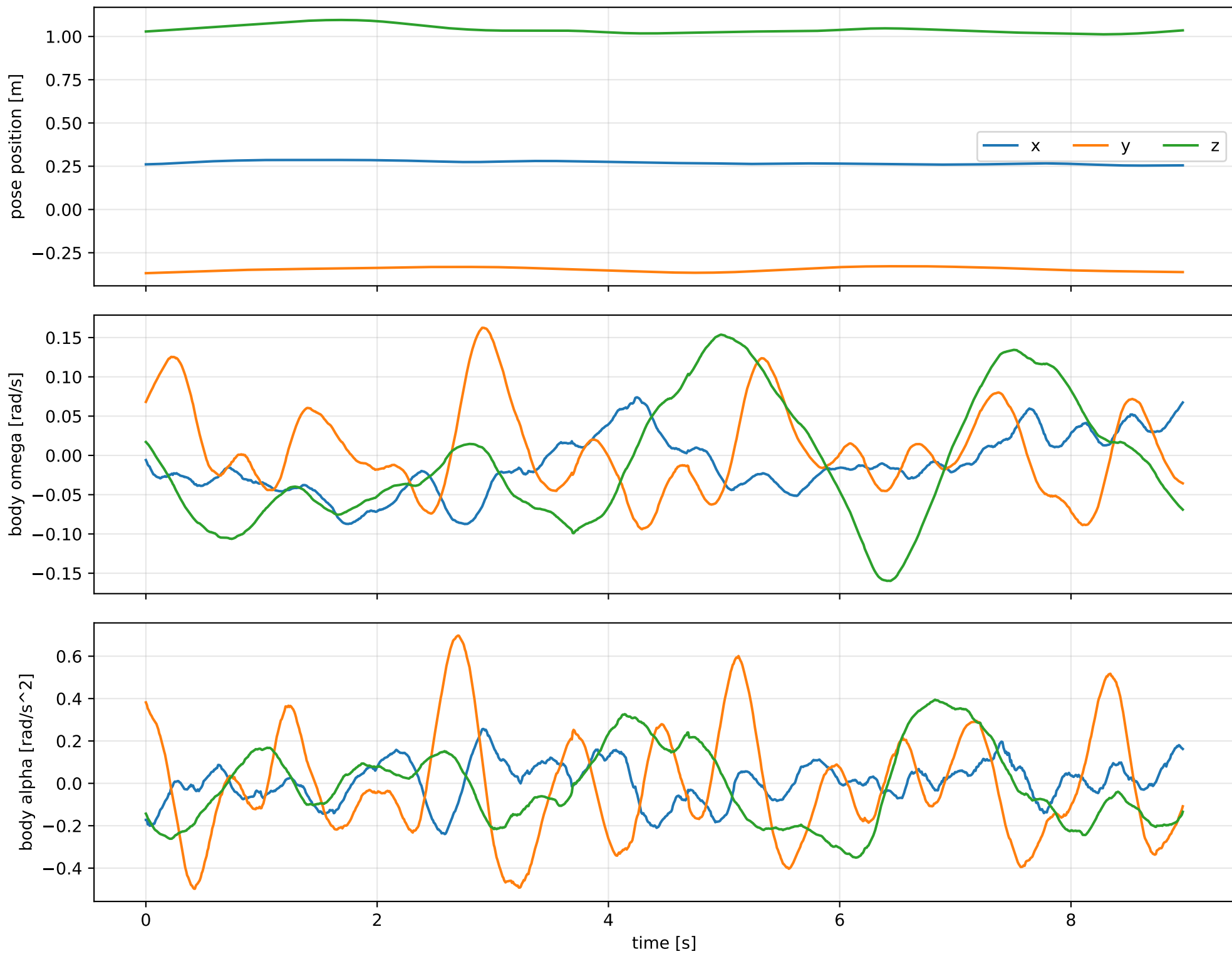
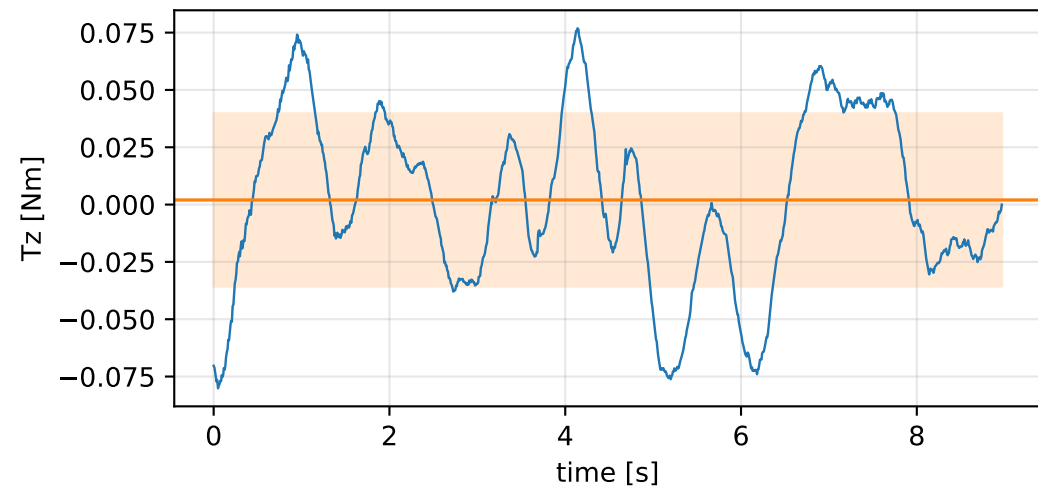
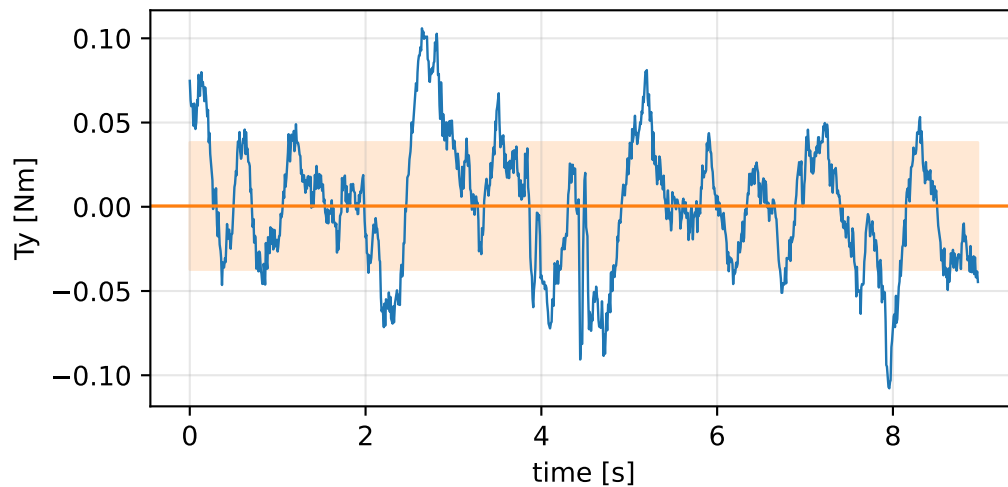
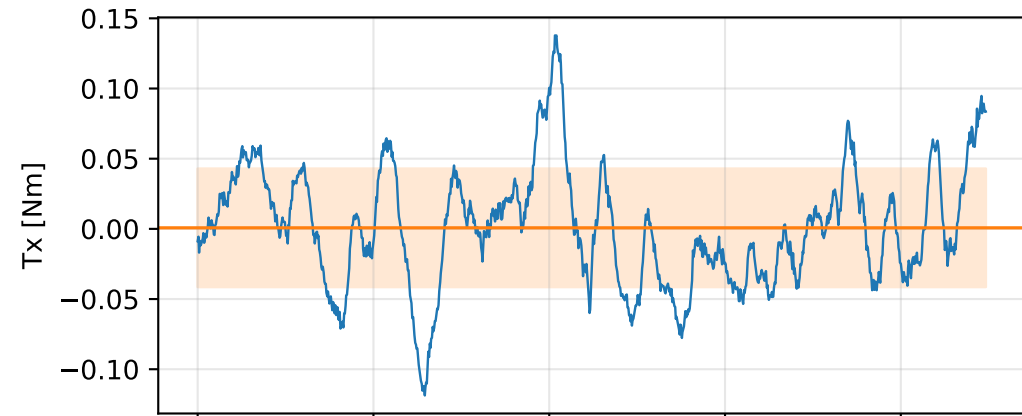
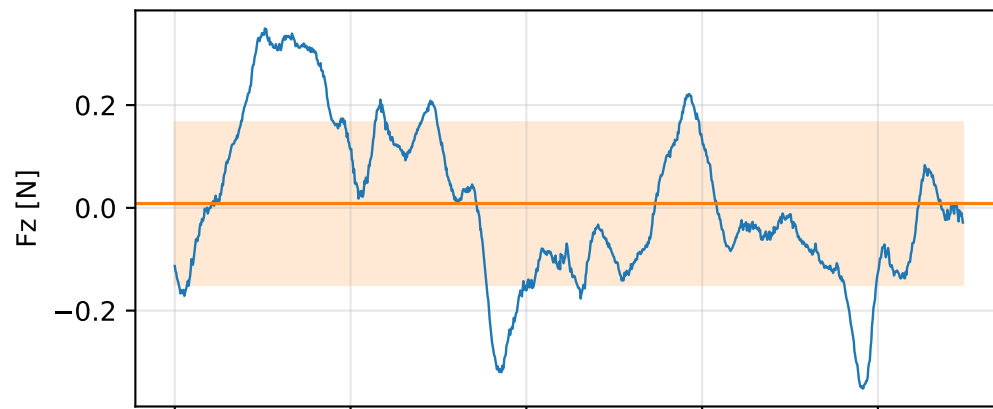
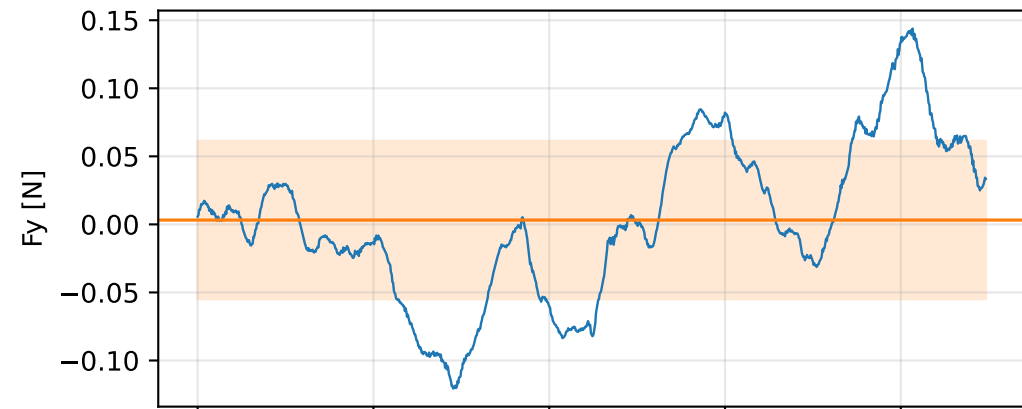
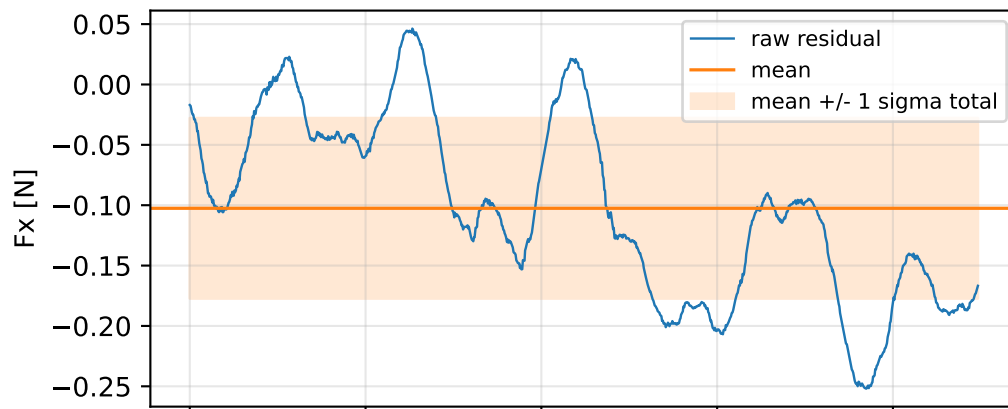


cog_z_nominal: SG trajectory and derived motion



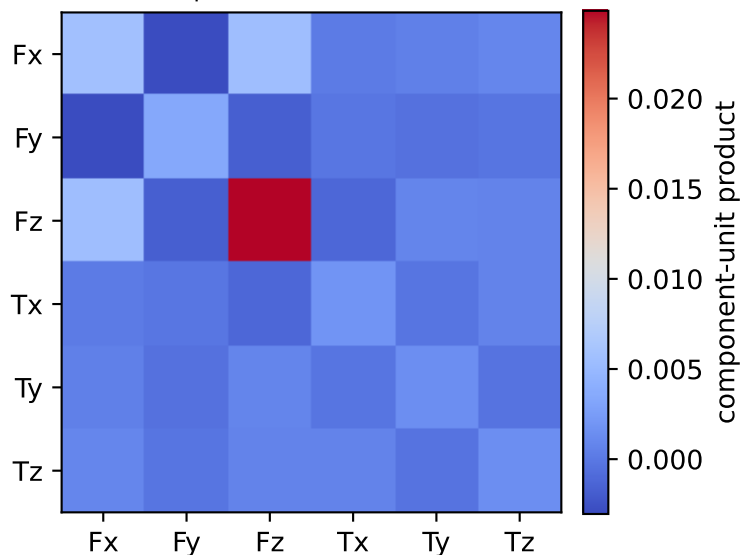
cog_z_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



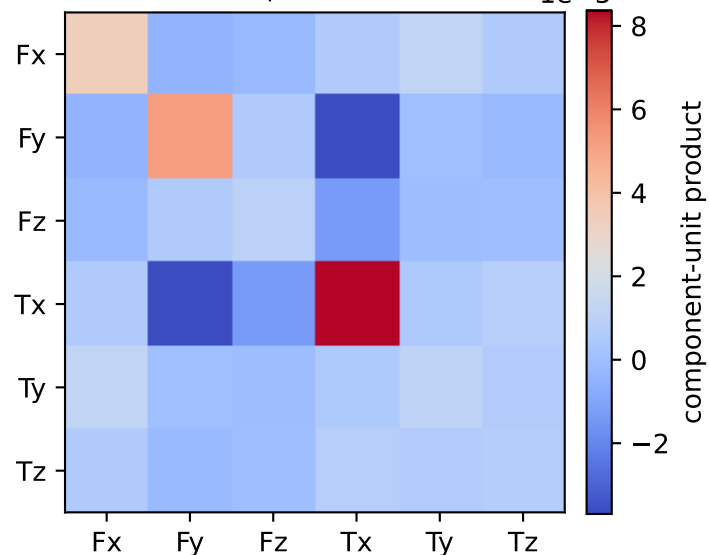
vector RMS about zero: force 0.21068 N, torque 0.067966 Nm; component std about mean: force [0.0747 0.058 0.1577], torque [0.0421 0.0377 0.0377]

cog_z_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.002 0.006 0.027]

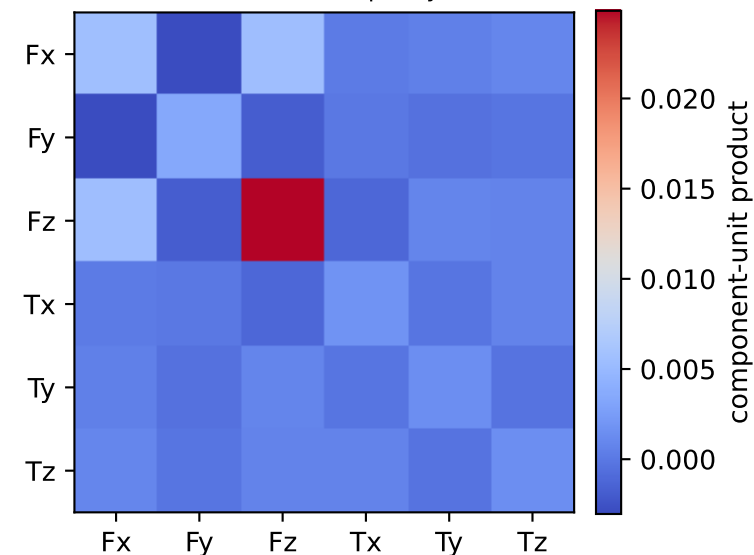
empirical total covariance



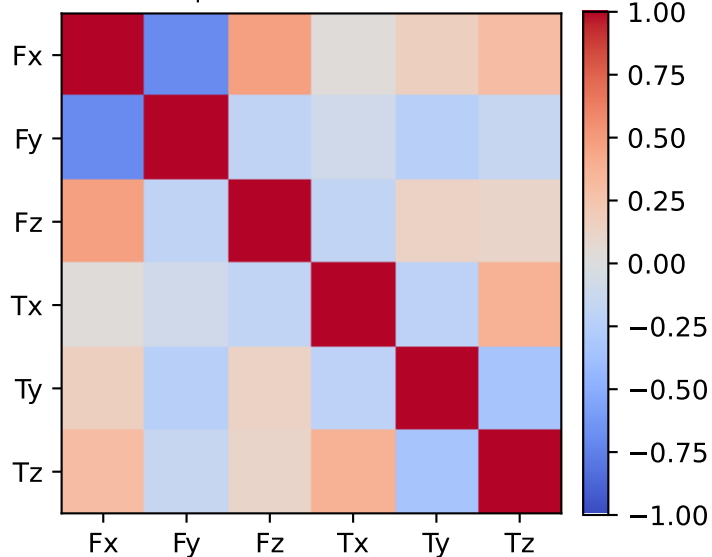
mean full-SG prediction covariance $1e-5$



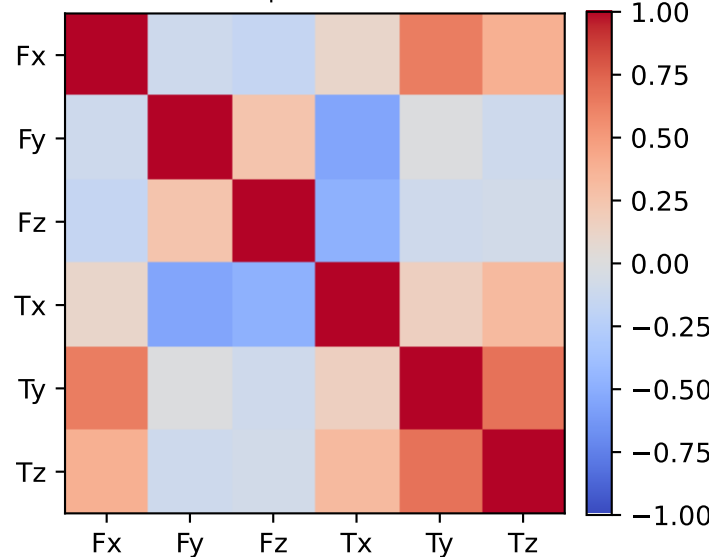
PSD excess model discrepancy covariance



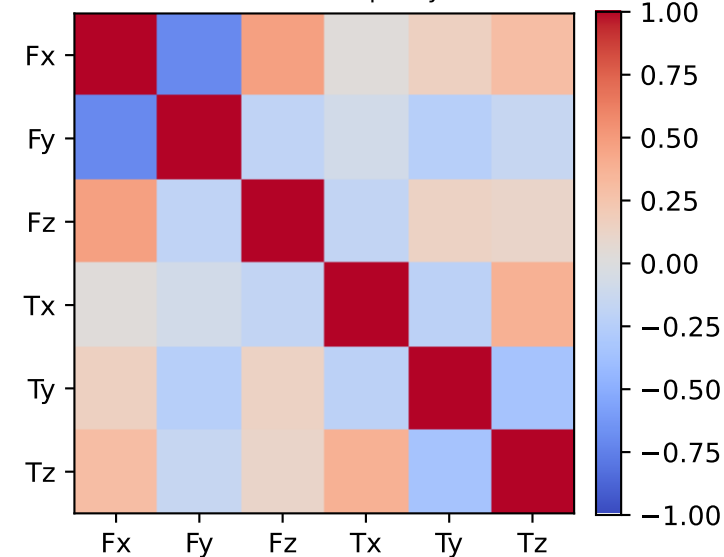
empirical total correlation



mean full-SG prediction correlation

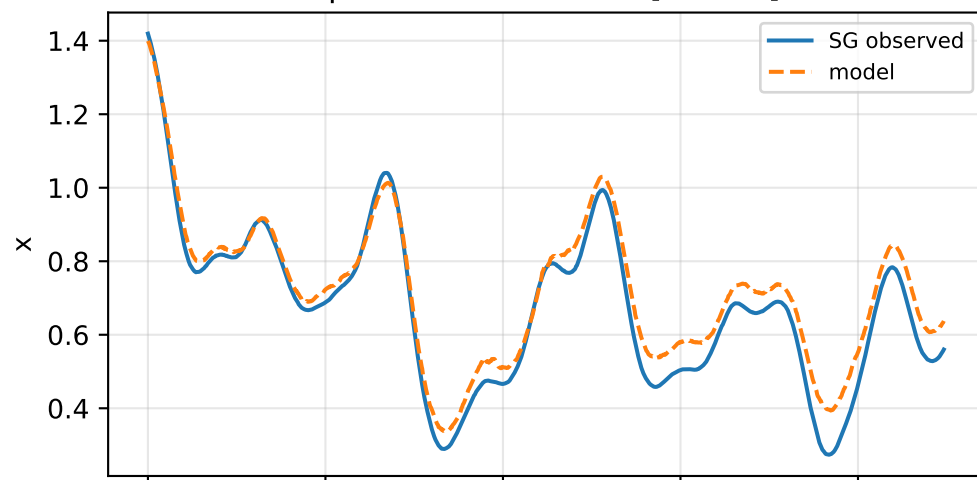


PSD excess model discrepancy correlation

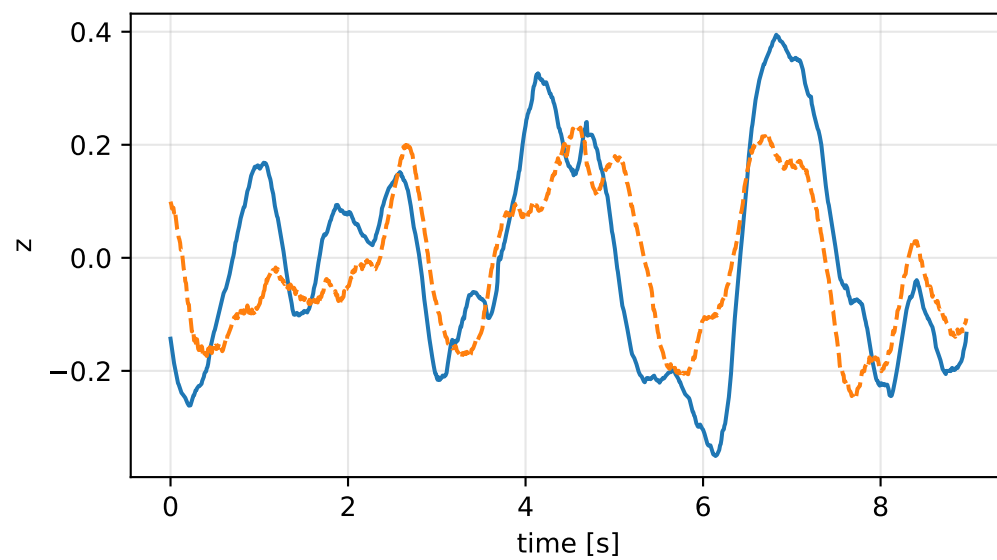
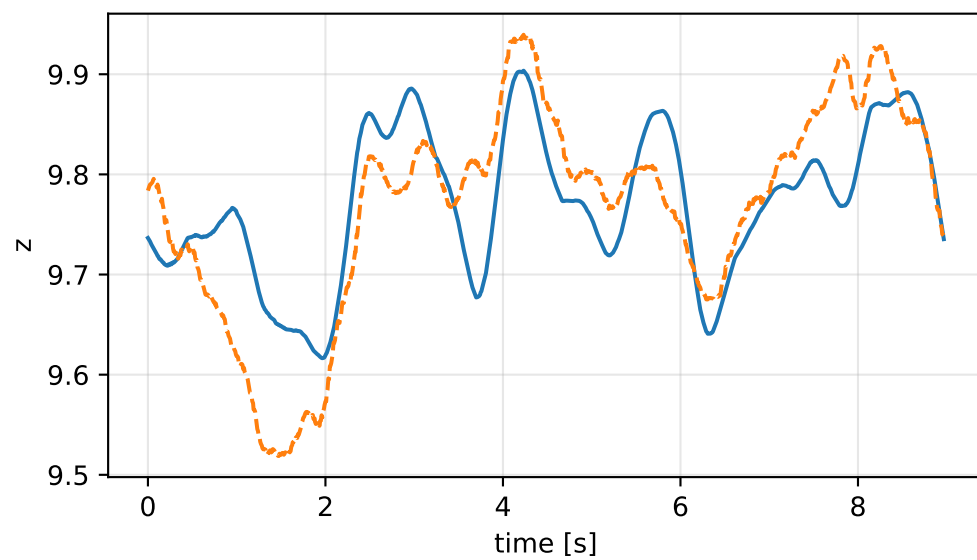
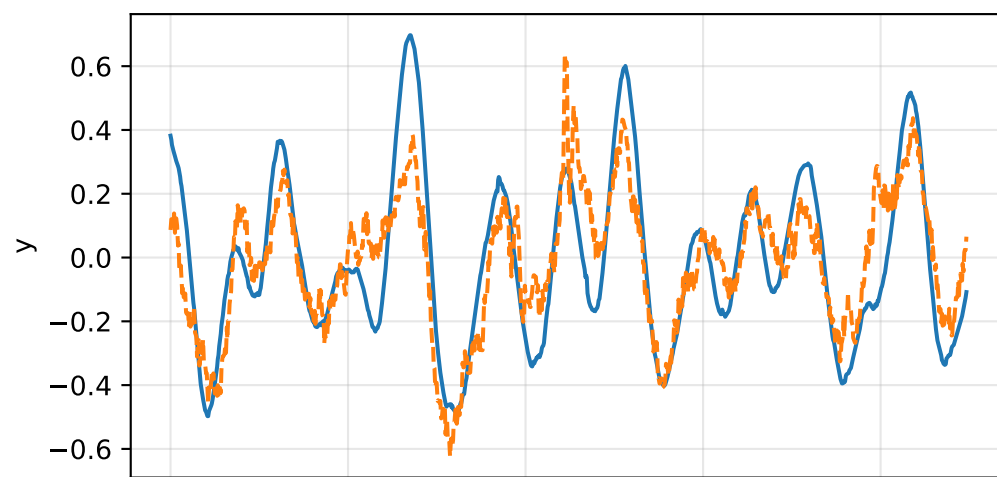
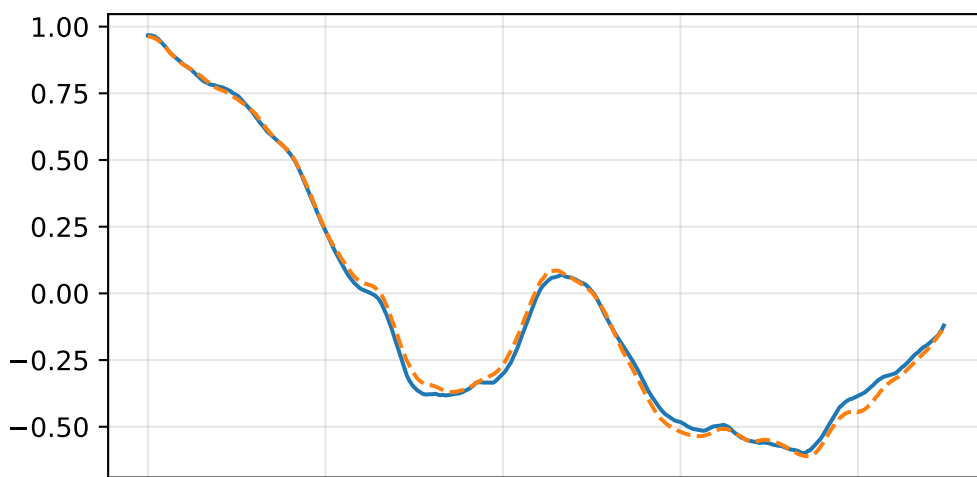
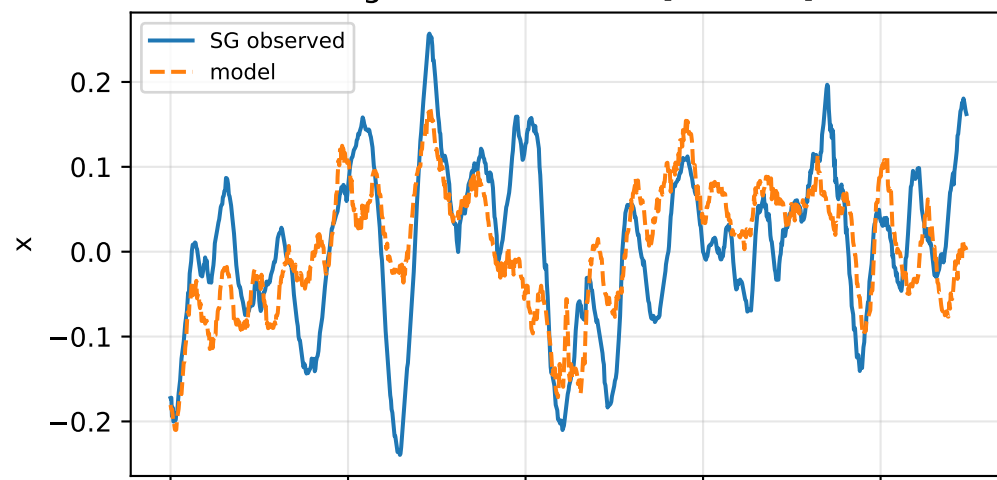


cog_z_nominal: acceleration residual objective

specific acceleration [m/s²]

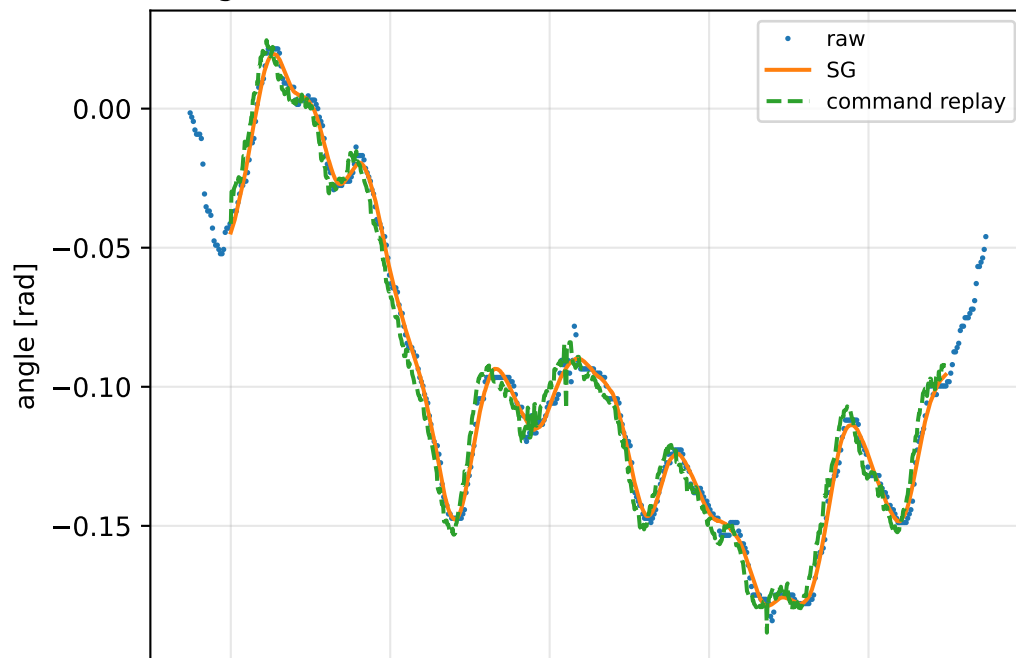


angular acceleration [rad/s²]

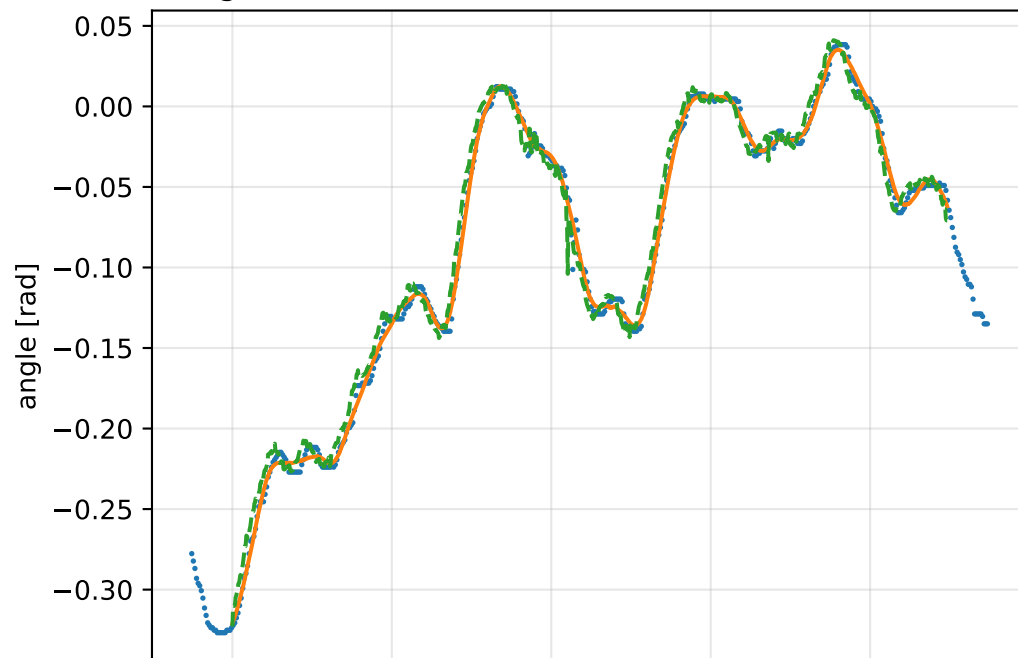


cog_z_nominal: gimbal measurement audit (objective=measured_sg)

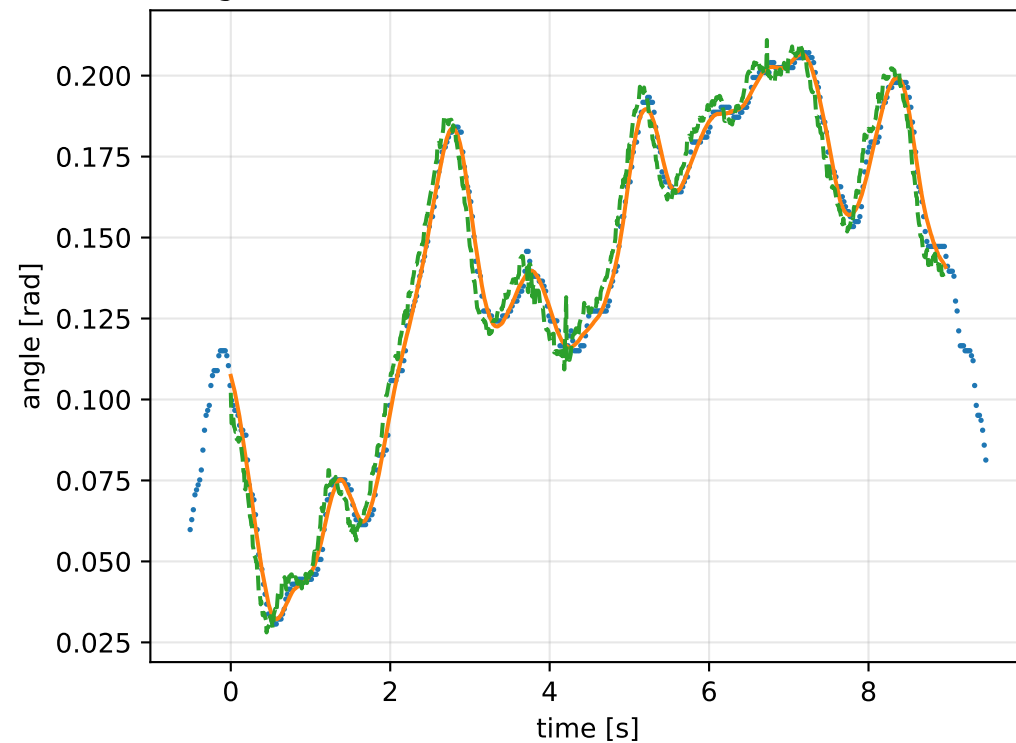
gimbal 1: RMSE=0.006345, max=0.01494 rad



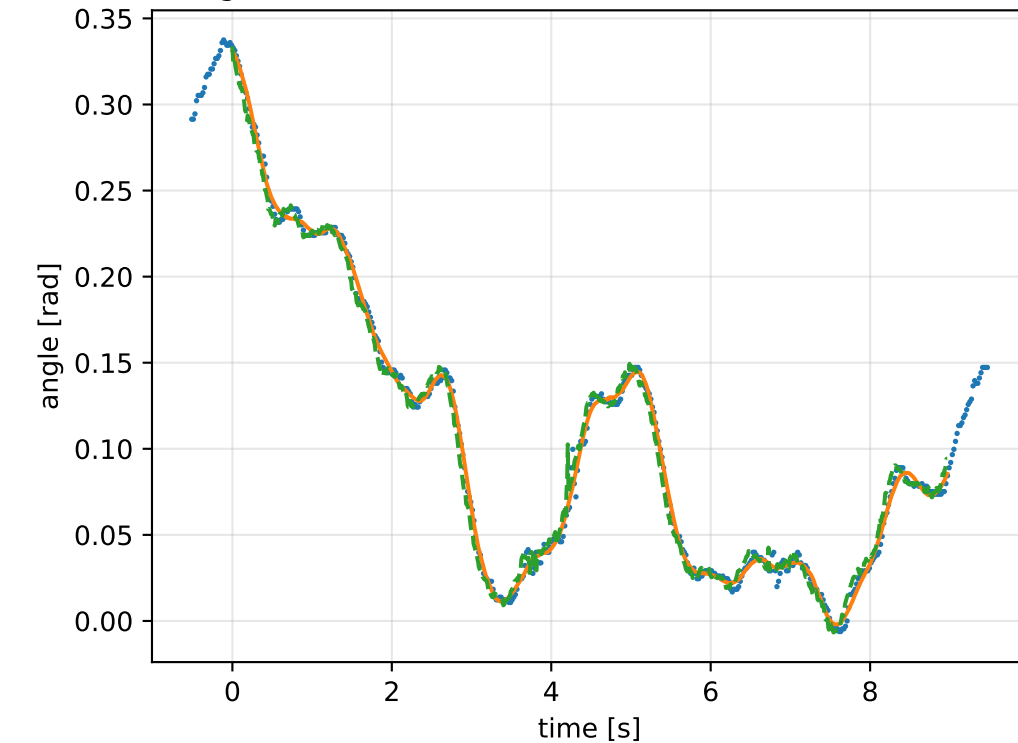
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

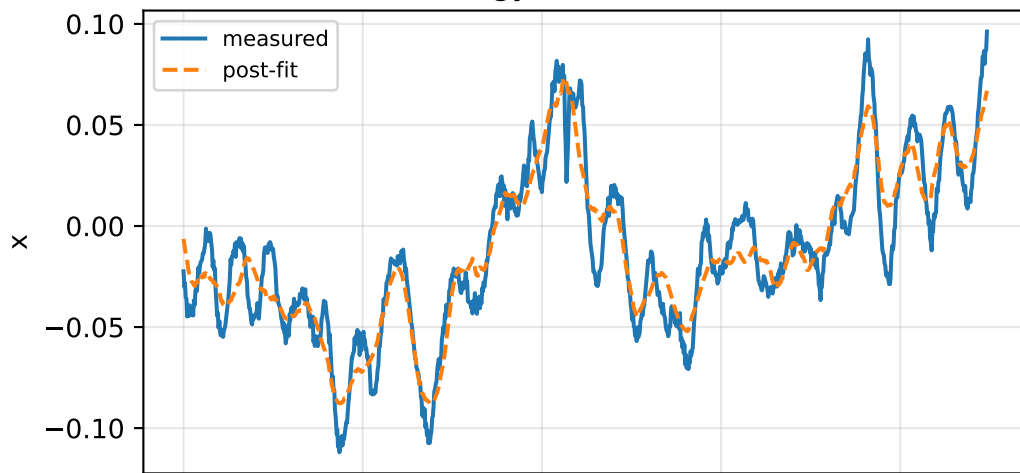


gimbal 4: RMSE=0.006321, max=0.03642 rad

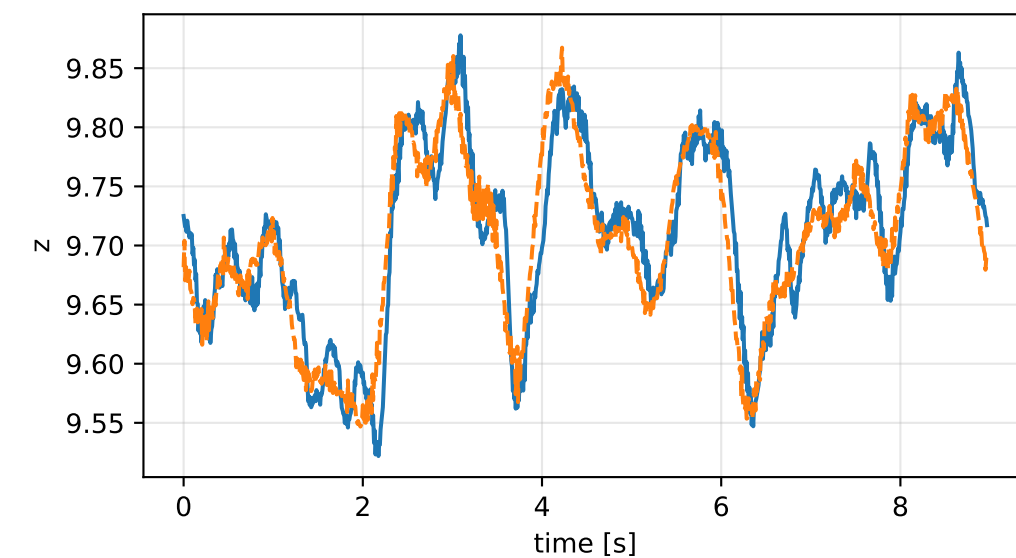
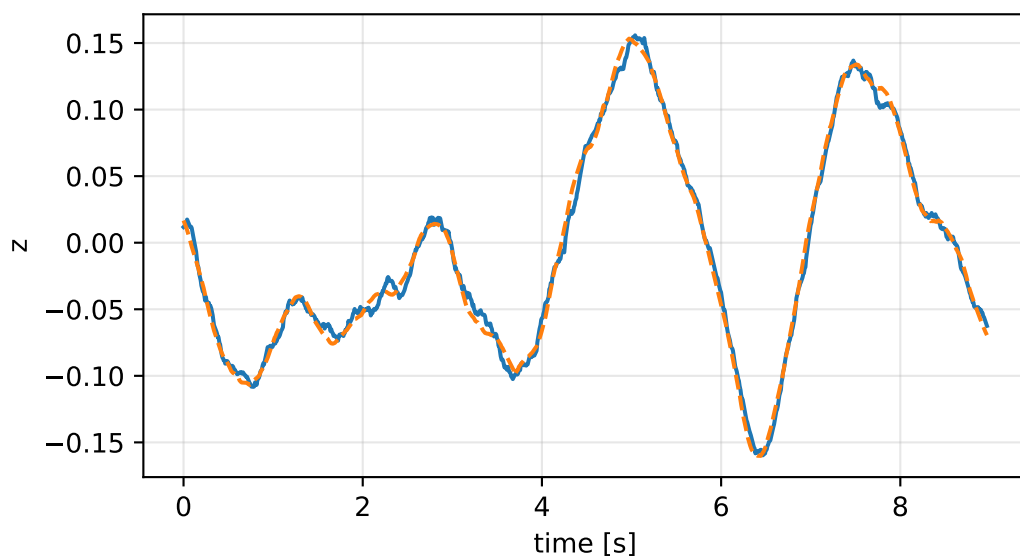
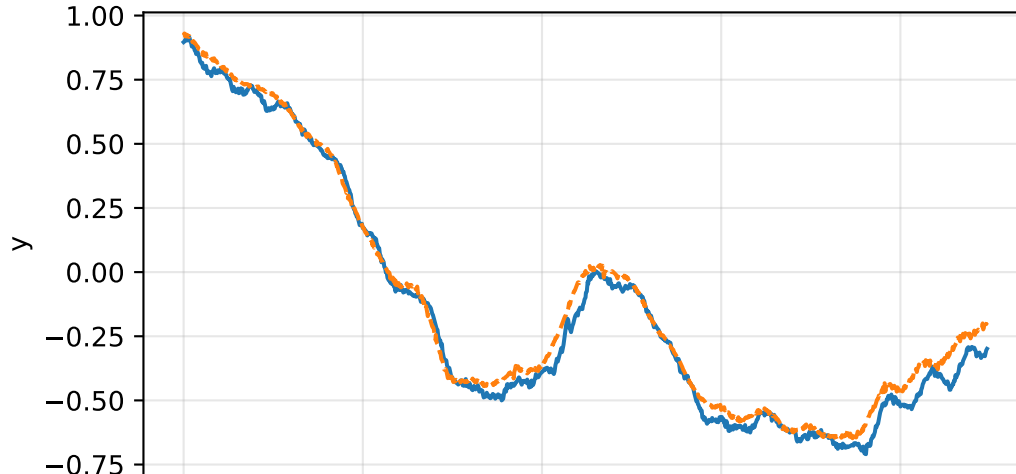
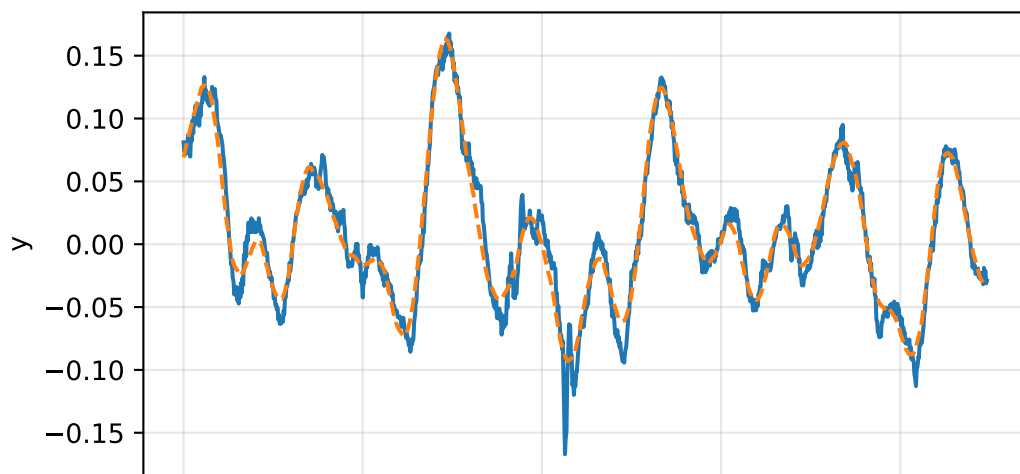
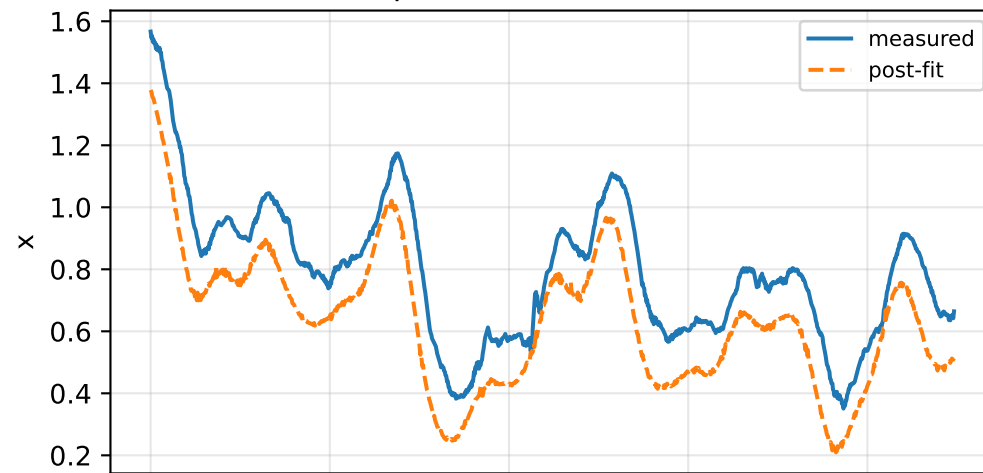


cog_z_nominal: IMU comparison (diagnostic only)

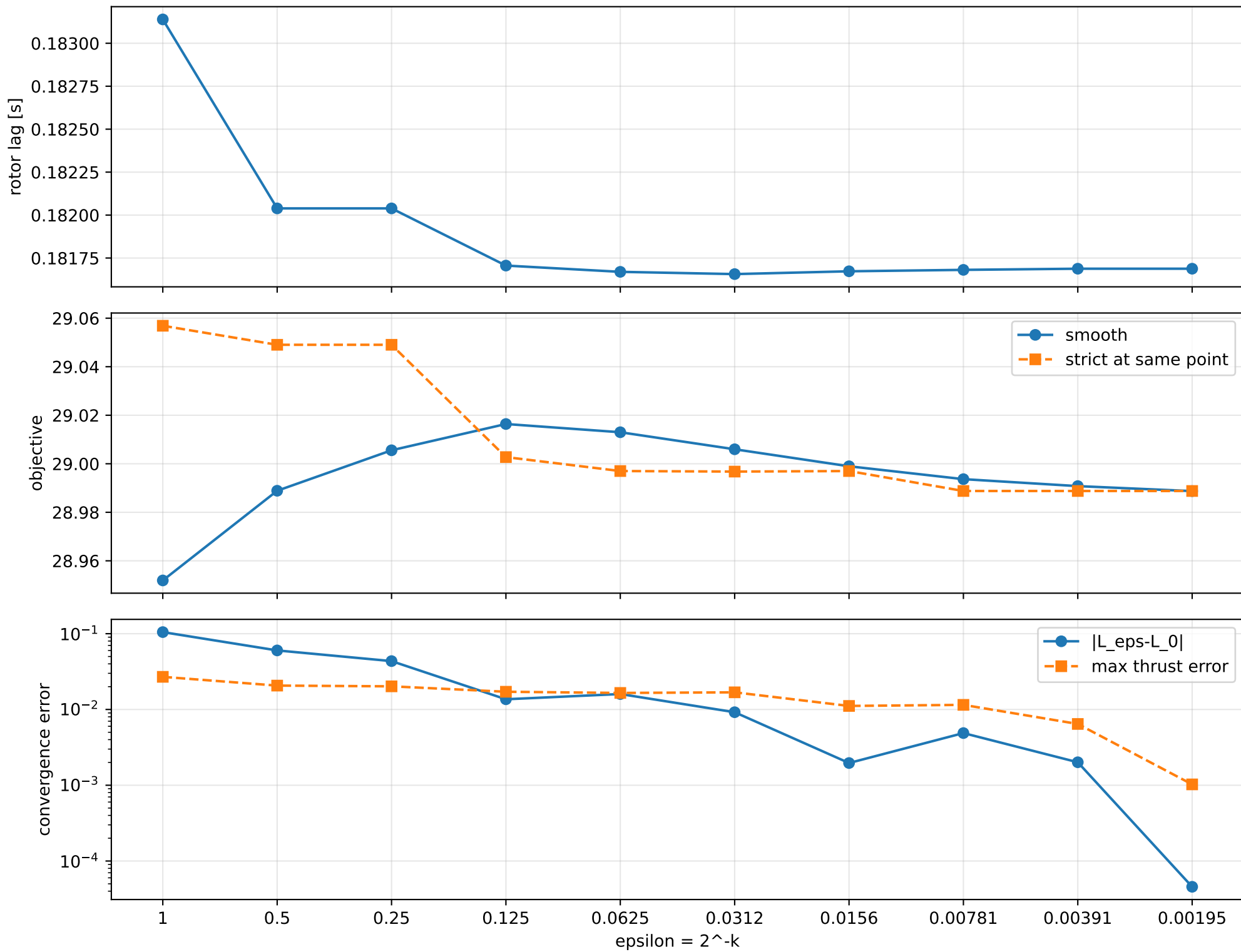
gyro [rad/s]



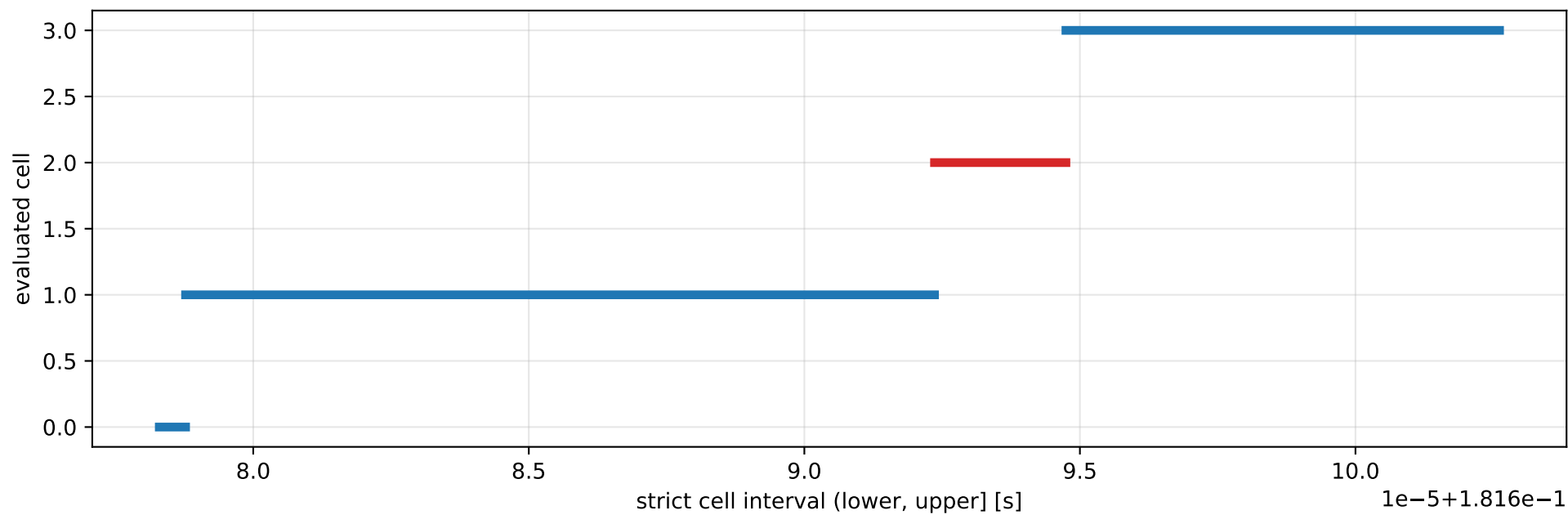
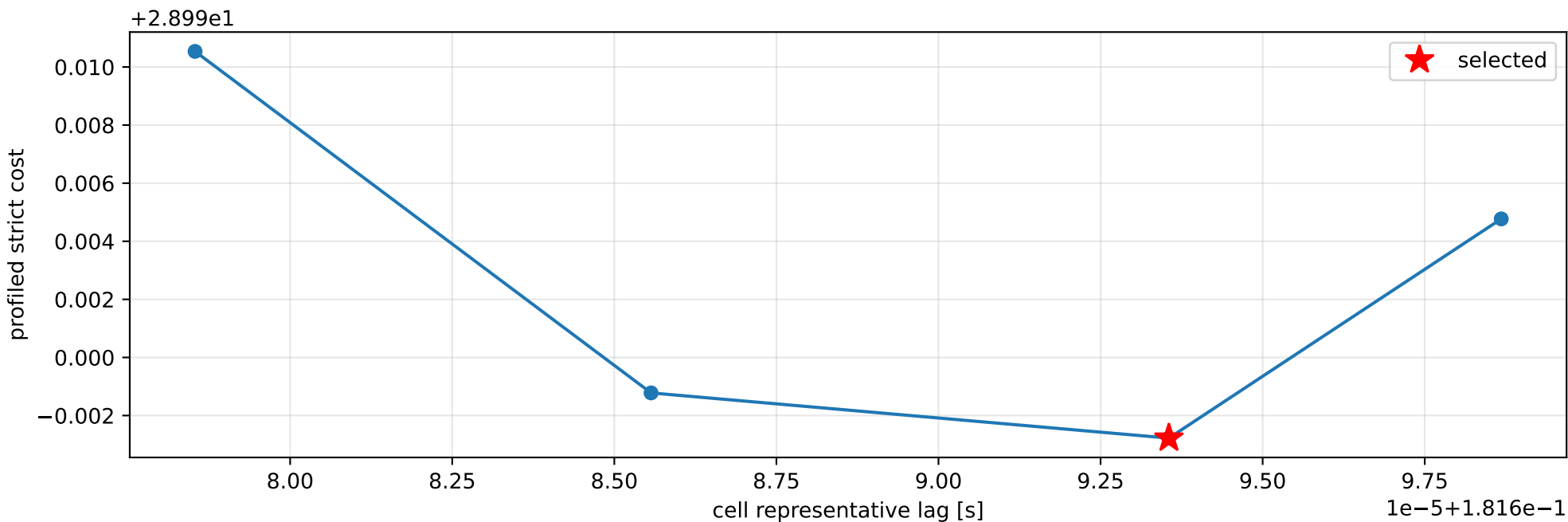
specific force [m/s^2]



cog_z_nominal: smooth-to-strict continuation



cog_z_nominal: exact strict-ZOH lag cells



selected (0.181692362, 0.181694746] s; final smooth 0.181687845 s; data boundary=False

cog_z_nominal: parameter estimate

Case: cog_z_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 43.9473855

inertia [kg m²]:

```
[[ 9.65296315 -0.20431736  0.7537417 ]  
 [-0.20431736  4.63221912 -0.12335496]  
 [ 0.7537417  -0.12335496  5.29311363]]
```

CoG body [m]: [-0.01582367 -0.05518201 0.00950974]

force effectiveness: [19.38825683 19.27452015 10.34462807 12.01789511]

scale-free J/m [m²]:

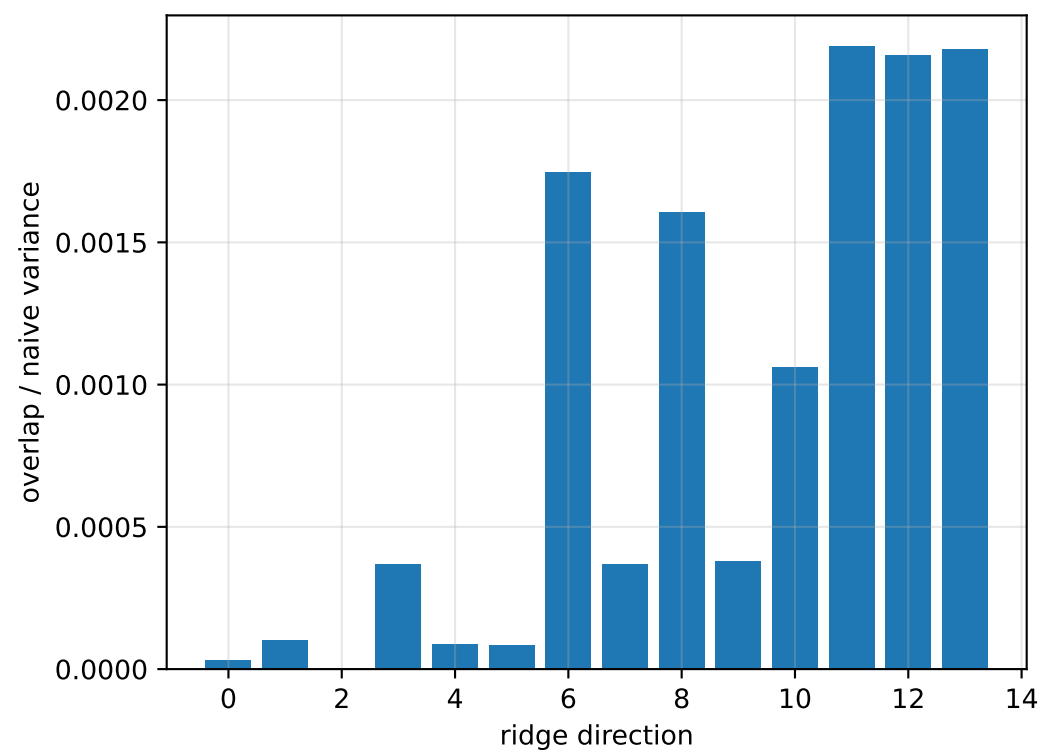
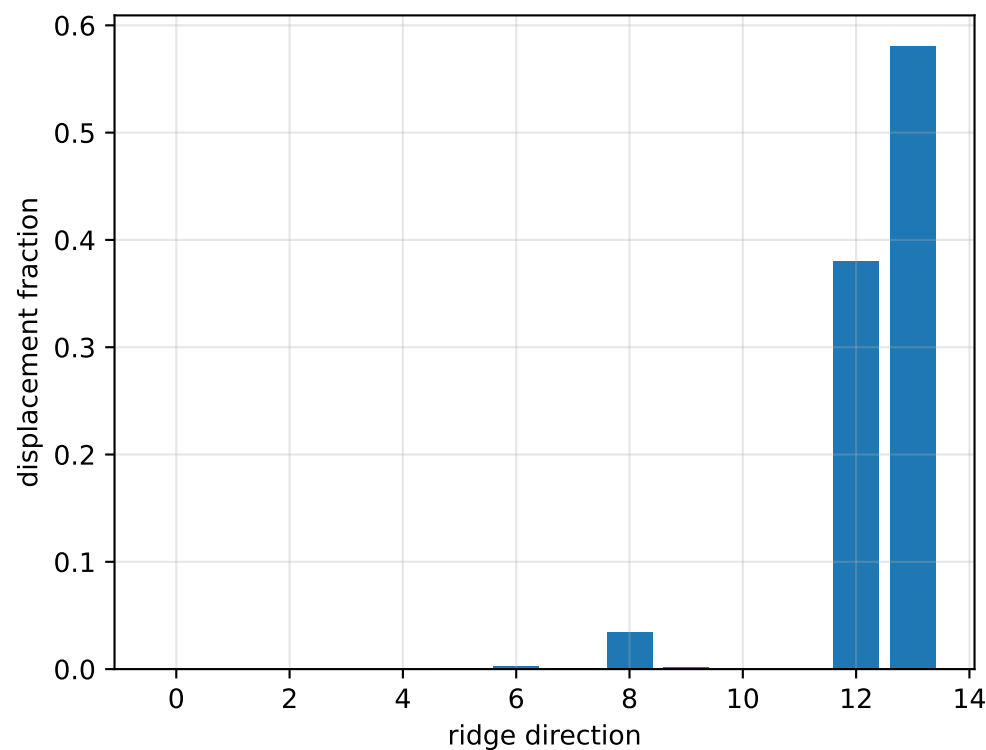
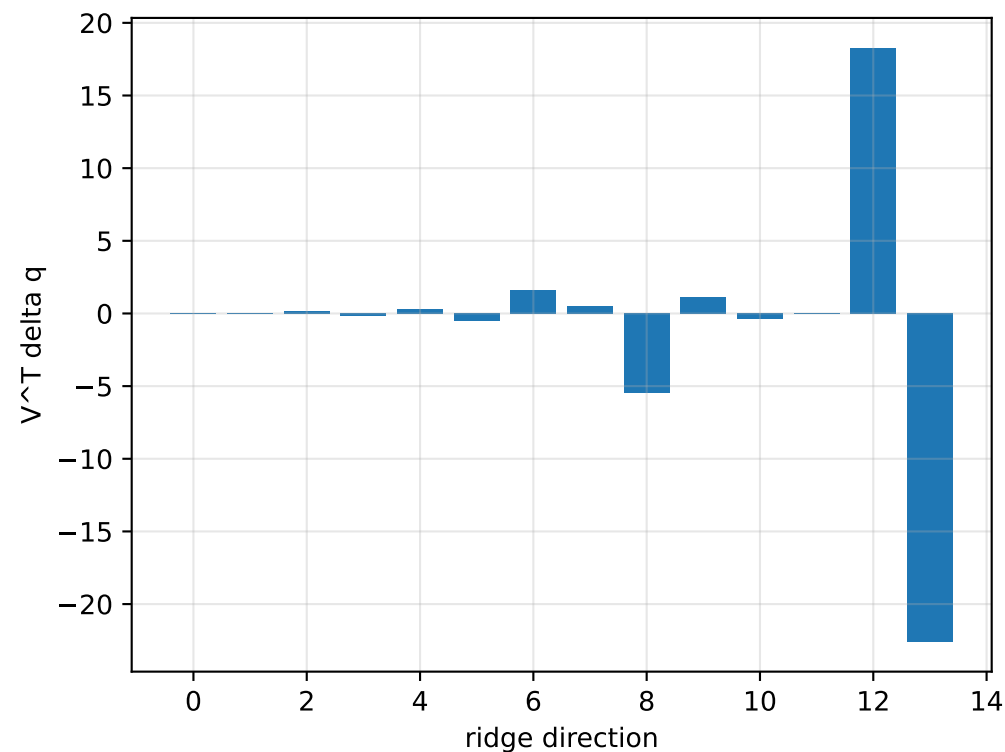
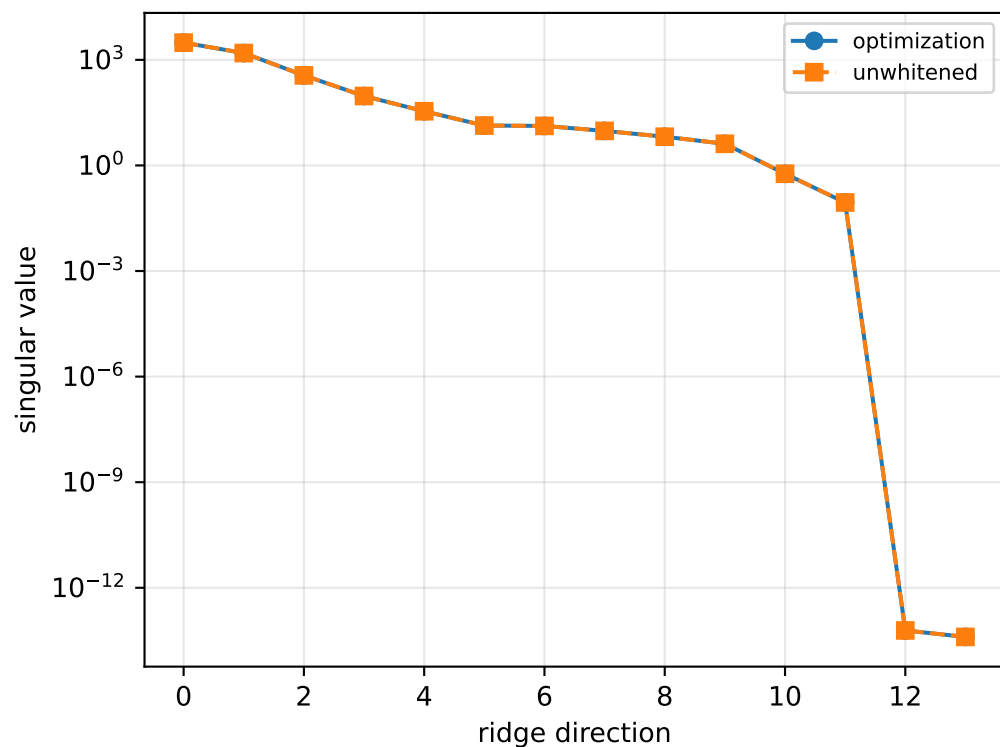
```
[[ 0.21964818 -0.00464914  0.017151 ]  
 [-0.00464914  0.10540375 -0.00280688]  
 [ 0.017151  -0.00280688  0.12044206]]
```

scale-free f/m: [0.44116974 0.43858173 0.23538666 0.27346098]

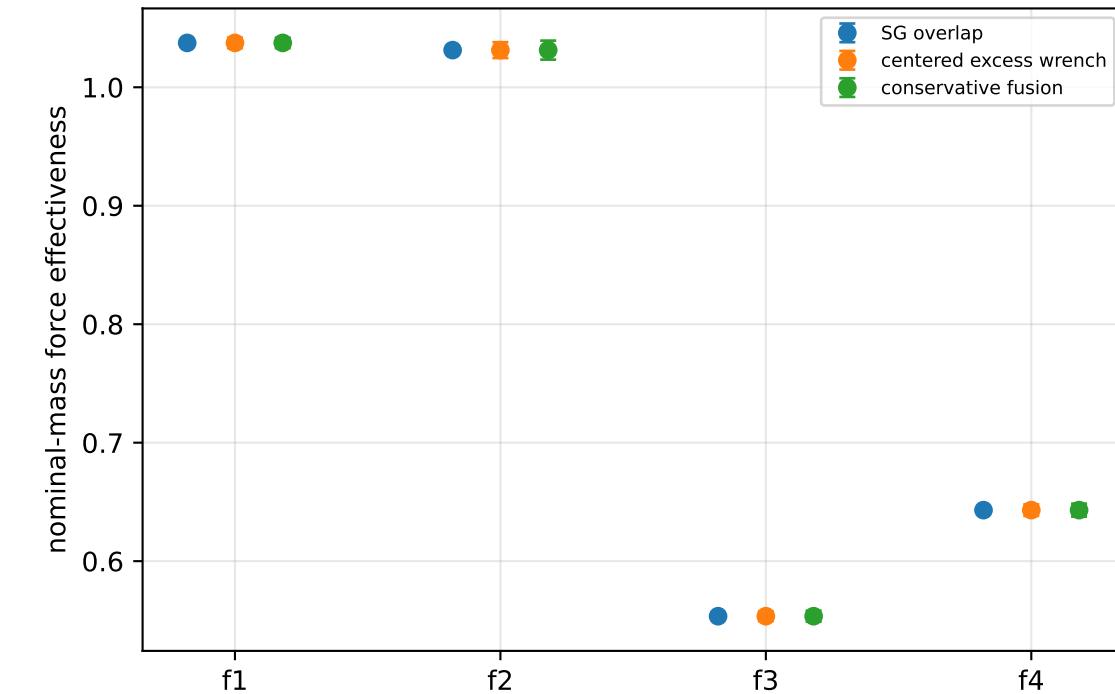
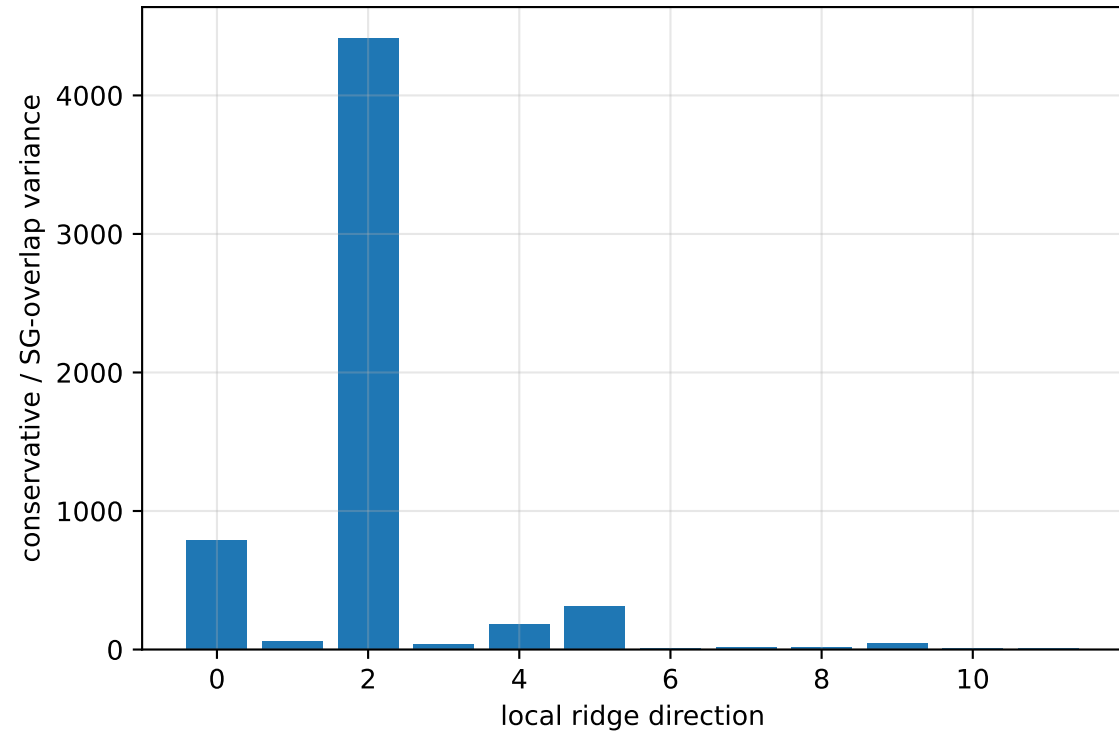
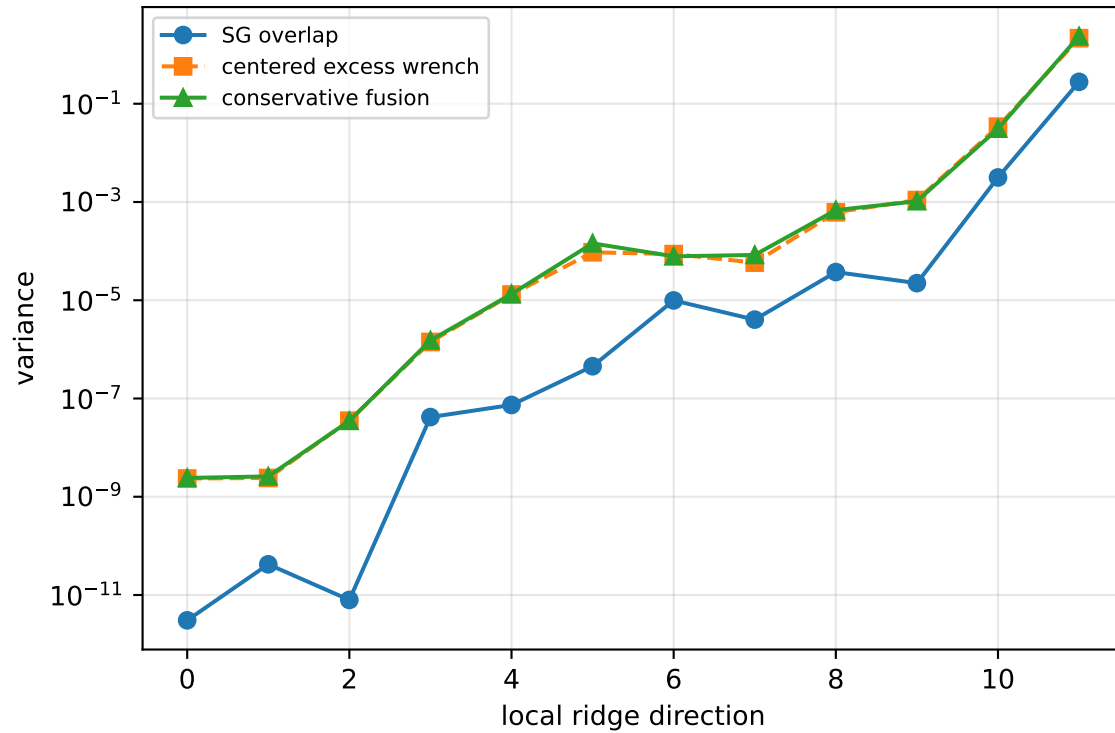
rotor lag cell: (0.181692362, 0.181694746] s

representative: 0.181693554 s

cog_z_nominal: ridge spectrum (rank 12, nullity 2, ||jv||=8.09e-14)



cog_z_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 442.5$
wrench-to-acceleration recovery max error = $2.85\text{e-}15$
exact scale gauge ridge index = 12 (alignment 0.777302)

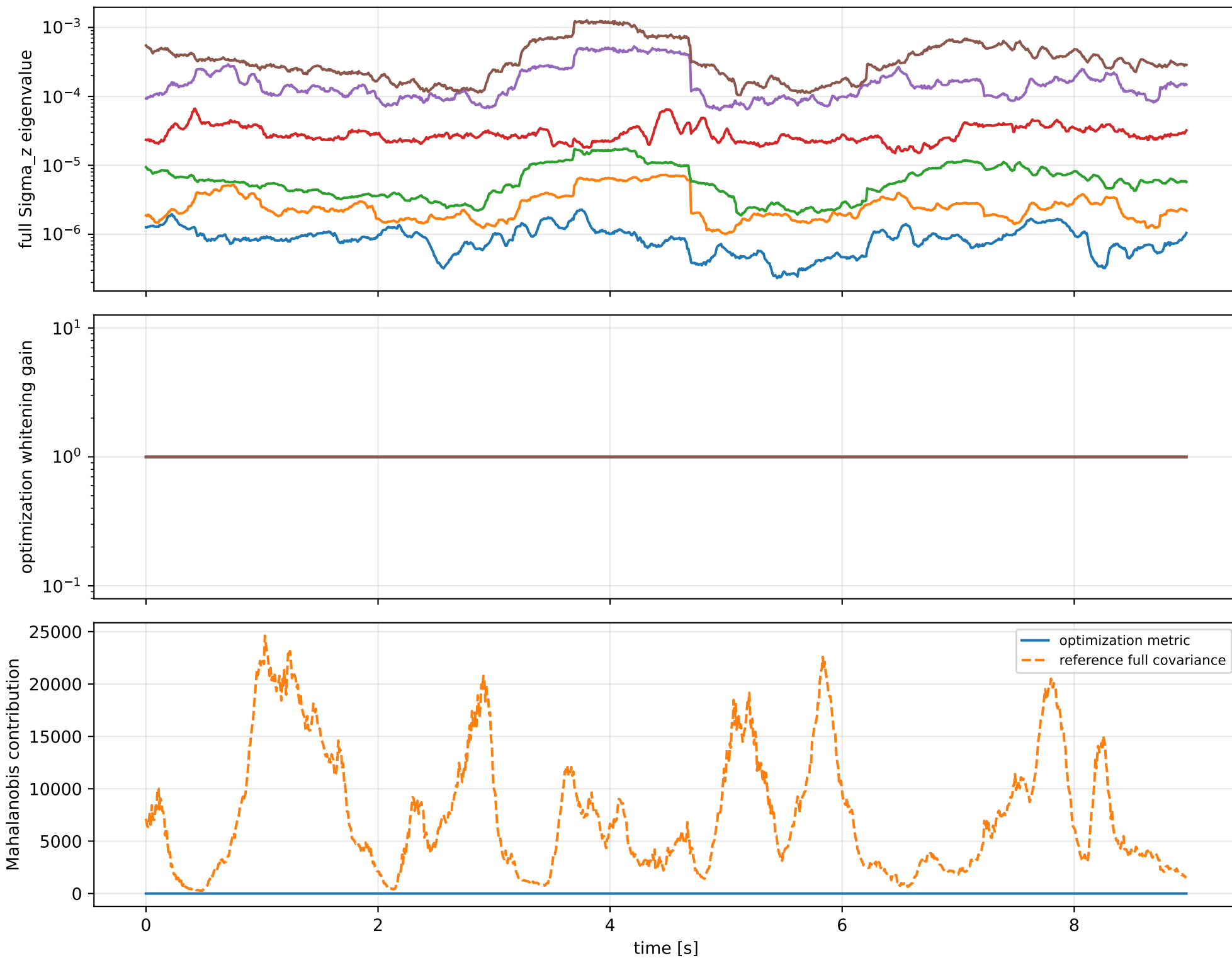
index 11: variance=2.364, ratio=8.463
 $\log_{\text{mass}}=+0.000864$; second_moment_diag_1=+0.0629
second_moment_diag_2=-0.061; second_moment_diag_3=-0.00668
second_moment_offdiag_12=+0.994; second_moment_offdiag_13=-0.0686
second_moment_offdiag_23=+0.0196; cog_x=-0.000114
cog_y=+0.000104; cog_z=-2.61e-05
log_force_eff_1=+0.00131; log_force_eff_2=-0.000425
log_force_eff_3=+0.00165; log_force_eff_4=+0.00135

index 10: variance=0.0311, ratio=9.845
 $\log_{\text{mass}}=+0.00995$; second_moment_diag_1=+0.0158
second_moment_diag_2=+0.0248; second_moment_diag_3=-0.0865
second_moment_offdiag_12=-0.0694; second_moment_offdiag_13=-0.992
second_moment_offdiag_23=+0.0406; cog_x=-0.00199
cog_y=+9.52e-05; cog_z=+2.96e-05
log_force_eff_1=+0.0154; log_force_eff_2=+0.00339
log_force_eff_3=-0.00555; log_force_eff_4=+0.0226

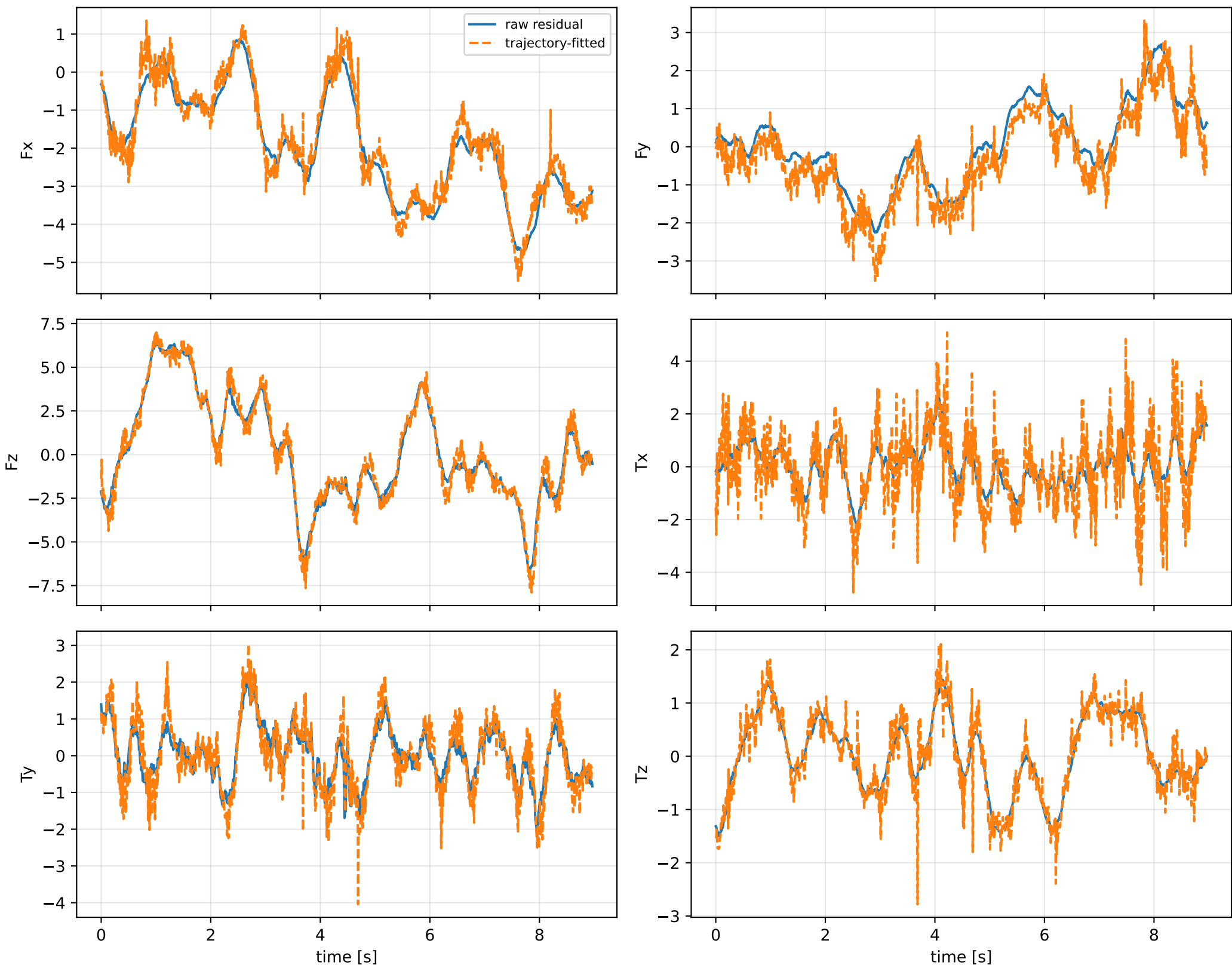
index 9: variance=0.001037, ratio=46.51
 $\log_{\text{mass}}=-0.165$; second_moment_diag_1=+0.00124
second_moment_diag_2=+0.896; second_moment_diag_3=-0.0594
second_moment_offdiag_12=+0.0529; second_moment_offdiag_13=+0.0257
second_moment_offdiag_23=+0.217; cog_x=-0.00148
cog_y=-0.00152; cog_z=-0.00482
log_force_eff_1=-0.166; log_force_eff_2=-0.153
log_force_eff_3=-0.202; log_force_eff_4=-0.151

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_z_nominal: optimization vs reference covariance



cog_z_nominal: wrench / closure (force max $1.25\text{e-}13$, torque max $8.4\text{e-}16$)



cog_z_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_z_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_z_nominal.j

SHA256: 0fc7d4f0f59d87c49bab62593db16eefada27449488e6f94ba9dea5cb570cbdb

data objective: 28.9872223441

prior objective: 4.24193980067e-07

total objective: 28.9872227683

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_z_nominal (cog_position_body_m)

components: z

target: [0.00950975]

std: [1.e-05]

final: [0.00950974]

error: [-9.21079779e-09]

standardized residual: [-0.00092108]

factor objective: 4.24193980067e-07